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Progressive knowledge tracing: Modeling learning process from abstract to concrete

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ABSTRACT

Artificial intelligence has the potential to revolutionize education by providing personalized learning experiences that support the dream of "teaching students according to their aptitude". Knowledge tracing (KT) is a critical research topic in intelligent education and is a powerful tool for achieving AI-assisted education. Both the learning process and the response result are important for KT. However, existing KT methods rarely employ a stage-based modeling approach to dissect the learning process. We propose Progressive Knowledge Tracing (PKT), which models the learning process in stages. PKT decomposes the learning process into three relatively independent but progressively related stages: concept mastery, question solving, and answering behavior. Inspired by constructivist learning theory and item response theory, PKT incorporates interpretable parameters with educational significance. Compared with existing KT methods, this staged modeling method that integrates educational theory has more reasonable interpretability. Experiments on six real-world datasets demonstrate that PKT outperforms baseline methods. Several experiments show that PKT reasonably models the learning process. For example, it more reasonably estimates the trend of concept mastery over time, analyzes the reasons why learners make mistakes on specific questions, and provides estimates of question difficulty that are closer to reality. We also find the intuitive phenomenon that the difficulty of a question is positively correlated with the number of associated concepts.

1. Introduction

Artificial intelligence technology is booming and flourishing rapidly. The emergence of GPT-4 signifies a breakthrough in AI technology, which will better penetrate and serve various fields. Exploring how to empower education with AI and help achieve the millennium dream of "teaching students according to their aptitude" (Yang, 2019) is an extremely attractive question. In addition, traditional classroom teaching has been challenged by the impact of the COVID-19 pandemic, leading to unprecedented attention being given to various online learning systems such as Coursera, ASSISTment, and Massive Open Online Courses (MOOCs). Online learning systems have accumulated vast amounts of student learning data, and how to extract potential learning patterns from this massive data to provide personalized guidance is a question that researchers are closely investigating (Liu, Shen, et al., 2021; Ramírez-Montoya, Andrade-Vargas, Rivera-Rogel, & Portuguez-Castro, 2021).

Knowledge tracing (KT) aims to track students' knowledge status and predict their future performance (Abdelrahman, Wang, & Nunes, 2022; Liu, Zou, et al., 2021; Sun et al., 2022). It is one of the most fundamental and critical research issues in intelligent education and a powerful tool for realizing AI-assisted education. Early knowledge tracing models such as Bayesian Knowledge Tracing (BKT) (Corbett & Anderson, 1994) and Performance Factor Analysis (PFA) (Pavlik Jr, Cen, & Koedinger, 2009) are characterized by their interpretability and educational significance in the modeling process. However, compared to deep learning models, their predictive accuracy may have certain limitations. Benefiting from deep learning techniques, Deep Knowledge Tracing (DKT) (Piech et al., 2015) has demonstrated superior performance in comparison to probability models such as BKT and logical models such as PFA. However, DKT suffers from a lack of interpretability, as it is unable to explicitly determine a student's level of knowledge mastery from hidden states.

When the performance is similar, interpretable models are more likely to gain the trust of teachers and students and effectively assist them. Especially in the field of education, one should know what works and why it works. People expect machines to not only provide learning

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suggestions but also explain why such suggestions are made. Otherwise, teachers and students cannot fully trust the advice given by the model. Therefore, research on the interpretability of deep knowledge tracing models is crucial to their practical implementation in teaching.

Many KT models have made efforts to improve their interpretability. The Dynamic Key–Value Memory Network (DKVMN) (Zhang, Shi, King, & Yeung, 2017) models introduces attention mechanisms and uses an external memory matrix to represent the learner's knowledge. Attentive Knowledge Tracing (AKT) (Ghosh, Heffernan, & Lan, 2020) proposes a novel monotonic attention mechanism that leverages an exponential decay curve to effectively down-weight the importance of questions in the distant past. Additionally, AKT utilizes Rasch model (Lord, 1980; Rasch, 1993) embeddings to capture individual differences between questions. These features endow AKT with both strong interpretability and performance, establishing it as a new benchmark for KT methods. Structure-based knowledge tracing (SKT) (Tong et al., 2020) considers one of the educational theories, knowledge transfer, and utilizes the relationships within the knowledge structure to model the spread of influence between concepts to improve the model's effectiveness and interpretability. Additionally, there are models that combine KT with some static cognitive diagnostic models, such as IRT-KT (Khajah, Huang, González-Brenes, Mozer, & Brusilovsky, 2014), Deep-IRT (Yeung, 2019), and Deep-IRTw (Tsutsumi, Kinoshita, & Ueno, 2021).

However, while the above methods have made significant contributions to the field, they may have some limitations in terms of considering the learning process, which is crucial for KT tasks. Innovative studies such as Learning Process-consistent Knowledge Tracing (LPKT) (Shen et al., 2021) and HawkesKT (Wang et al., 2021) effectively demonstrate the importance of studying the learning process in Knowledge Tracing. LPKT evaluates students' knowledge states by modeling their learning process, capturing learning gains and considering the phenomenon of forgetting in the learning process. Meanwhile, HawkesKT pioneers the introduction of the Hawkes process into this domain to explore temporal cross-effects in KT. These advanced models have not only achieved breakthroughs in studying the learning process but have further confirmed the undeniable value of focusing on the learning process within knowledge tracing.

Unlike existing models that prioritize the temporal aspects of the learning process, our study focuses on the learning process through the lens of students' sequential thinking steps. It models the learning process in a progressive, step-by-step manner, providing learners with feedback throughout the process.

In the field of motor learning, feedback has been typically classified into Knowledge of Results (i.e. information relating to the movement outcome; KR) (Bilodeau, 1966; Newell, 1976) and Knowledge of Performance (i.e. information on the movement process; KP) (Gentile, 1972). Both types of learning feedback can impact learning efficiency (Oppici, Dix, & Narciss, 2021). In KT tasks, KR only tells us the student's answer results, and we need KP to understand how the student answered the question, to gain a more comprehensive understanding of the learning process, and provide targeted support and feedback. Motivated by this perspective, the primary objective of our paper is to delve into the cognitive processes students engage in when applying their knowledge. We aim to furnish learners with more granular, process-oriented insights that can aid them in identifying the root causes of their mistakes. To achieve this, we develop the Progressive Knowledge Tracing model that segments the learning process into distinct, interpretable stages. This innovative approach not only enhances the model's interpretability but also introduces a fresh research angle to the existing body of work in this field.

Concerning how to deconstruct the thinking process of students when doing practice exercises, Baddeley's working memory model (Baddeley, 2003) offers valuable insights. When faced with a new learning task or question, students initially engage in a preliminary, abstract level of thinking in the central executive, attempting to decode and encode information in their working memory's visuospatial sketchpad and phonological loop. This initial understanding is abstract, as it mainly involves a conceptual grasp and association of the new information, rather than delving into specific applications or practice. As learning advances, however, students begin to link these abstract concepts with prior knowledge and skills stored in long-term memory, eventually transforming them into concrete problem-solving strategies or applications. This not only reflects a deepening understanding of the concept during the learning process but also emphasizes the critical transition from abstract theoretical knowledge to concrete practical application.

As shown in Fig. 1, when a student encounters a question, he/she has a preliminary understanding of the question and quickly retrieves the corresponding concepts. Then, the student engages in a deeper analysis of the question to determine the most appropriate problem-solving strategy and method. At this point, the mastery levels of the corresponding concepts and the difficulty of the question both influence the degree to which the student can solve the question. After thinking and determining the question-solving strategy, the student begins to write answers. During the answering process, there may be some behaviors that do not match the student's true mastery of the question, such as slipping and guessing.

To model the learning process from abstract thinking to concrete answering behavior, we propose the Progressive Knowledge Tracing (PKT) method, which divides the learning process into three relatively independent but progressively advanced stages: concept mastery, question-solving, and answering behavior. In the stage where the student retrieves concepts, PKT captures the degree of concept mastery, which affects the retrieval results (e.g., a concept that is almost unmastered may cause the student to overlook it). PKT constructs the loss function of concept mastery degree based on the idea of active construction and insight in constructivist learning theory (Bruner et al., 1966; Cobb & Bowers, 1999; Dewey, 1910; Duffy & Jonassen, 2013; Perkins, 1981). In the question-solving stage, PKT estimates the difficulty of the question and models the dual influence of concepts and questions. Inspired by the two-parameter logic model (2PLM) (Birnbaum, 1968), the mastery of corresponding concepts, question difficulty and discrimination are constrained together to describe the student's ability to solve a question. In the answering behavior stage, PKT captures two random behaviors that interfere with the true mastery level (slipping and guessing) and predicts whether the student can answer the next question correctly by integrating the question-solving ability and answering behavior.

Existing KT methods seldom model the learning process by dividing it into these three stages and comprehensively capturing each stage. As shown in Table 1, the classical BKT probability model considers both the mastery of concepts and answering behavior. Deep learning models such as DKT, DKVMN, and SKT model the student's mastery of concepts in different ways. Inspired by IRT (Lord, 1980), AKT, Deep-IRT, and Deep-IRTw not only consider the mastery of concepts, but also utilize question information for modeling. PKT completely models the three phases of the learning process.

To a certain extent, PKT reproduces the learning process and holds valuable potential for interpretability and practical application. By gradually modeling students' learning process from abstract to concrete, PKT can effectively capture the complex interactions among the students' mastery of concepts, solving of questions, and their answering behaviors, resulting in accurate and interpretable predictions. By analyzing how students solve questions, PKT can estimate the areas in which they may have misconceptions or gaps in understanding. This enables educators to better understand the students' cognitive states and helps intelligent tutoring systems provide personalized feedback or interventions to improve their performance. Based on the above analysis, our key contributions are as follows.

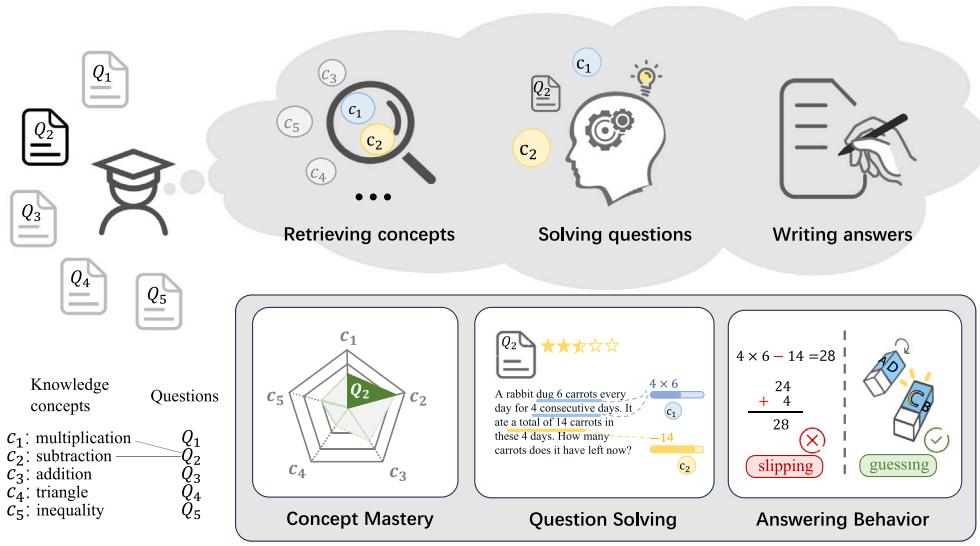

Fig. 1. Breaking down the student's learning process. The upper half of the figure shows a series of progressively advancing behaviors that occur during a student's progression from encountering a question to completing the answer. The lower half of the figure corresponds to the modular modeling of each step. During the concept retrieval phase, the model constructs the level of mastery of corresponding concepts; during the question-solving phase, the model takes into account the characteristics of the question and the level of the student's mastery of the concept; during the answer writing phase, the model considers the random answering behaviors that the student may exhibit.

Table 1
Comparison of the modeling process between the existing KT methods and PKT.

	BKT	DKT	DKVMN	SKT	AKT	Deep-IRT	Deep-IRTw	PKT
Concept Mastery	✓	✓	✓	✓	✓	✓	✓	✓
Question Solving					✓	✓	✓	✓
Answering Behavior	✓							✓

(1) Proposing the Progressive Knowledge Tracing (PKT) method, which decomposes the learning process into three progressively advanced stages. To our knowledge, this approach has not been previously explored in the literature.

(2) PKT progressively models the learning process from abstract to concrete, including from thinking to behavior of students, from concepts to questions. Furthermore, inspired by constructivist learning theory and item response theory, PKT designs a loss function with educational significance and generates interpretable parameters.

(3) Experiments on six public real-world datasets demonstrate that PKT outperforms baseline methods. Moreover, PKT reasonably models the learning process. The estimated trend of concept mastery aligns more closely to the actual learning process. The question difficulty learned by PKT is more accurate than that of Deep-IRT. Additionally, we find an intuitive phenomenon that the difficulty of a question is positively correlated with the number of associated concepts.

The code is available at https://github.com/WeiMengqi934/PKT.

2. Related works

2.1. Bayesian knowledge tracing

One representative model of such kind is the Bayesian Knowledge Tracing (BKT) (Corbett & Anderson, 1994) model, which uses a hidden Markov model to infer the latent state of knowledge mastery based on the observable states of whether students answer questions correctly. It uses four variables to model the learning process: the initial mastery probability $P(L_0)$; the probability of transition from the unlearned state to the learned state $P(T)$; the probability of guessing the correct answer in spite of non-mastery $P(G)$; and the probability of making a mistake in spite of mastery $P(S)$. Although it models the changes in students' knowledge states well, it does not consider the differences in students' abilities or the differences among questions. Many Scholars have made a lot of contributions to improving BKT, such as personalizing students' initial experiences (Pardos & Heffernan, 2010), considering forgetting factors (Qiu, Qi, Lu, Pardos, & Heffernan, 2011), taking into account item difficulty (Pardos & Heffernan, 2011), considering the similarity between items (Hawkins & Heffernan, 2014), using dynamic learning rate instead of the original static learning rate (Agarwal, Baker, & Muraleedharan, 2020), improving for binary variables (Falakmasir, Yudelson, Ritter, & Koedinger, 2015; Zhang & Yao, 2018), considering learners' emotions or mental state (Spaulding & Breazeal, 2015), etc.

2.2. Deep learning-based knowledge tracing

Deep learning has attracted extensive attention from researchers due to its powerful feature extraction capabilities, bringing new opportunities and challenges to the field of knowledge tracing. Many researchers have applied it to the KT field, known as deep learning-based knowledge tracing (DLKT).

In 2015, Piech et al. proposed for the first time the use of deep neural networks to construct a knowledge tracing model, namely Deep Knowledge Tracing (DKT) model (Piech et al., 2015) based on Recurrent Neural Networks (RNN). Compared with IRT and BKT, the DKT model significantly improves prediction accuracy. The development of KT models based on RNN has progressed rapidly (Lee & Yeung, 2019; Piech et al., 2015; Yeung & Yeung, 2018). However, the black-box nature of deep learning has led to DLKT models having varying degrees of interpretability and poor controllability. Nonetheless, scholars have been making efforts to improve the interpretability of DLKT models.

Dynamic Key–Value Memory Networks (DKVMN) (Zhang et al., 2017), which is based on key–value matrices for knowledge tracing. This model introduces a key matrix and a value matrix to respectively model concepts and students. Although it proposes an elegant method for modeling the process of concept learning, the vector representing the knowledge state still lacks interpretability. Some KT models consider educational theory such as Structure-based Knowledge Tracing (SKT) (Tong et al., 2020), Ability Boosted Knowledge Tracing

(ABKT) (Liu et al., 2022). The SKT model considers the knowledge transfer, one of the educational theories, and models the influence propagation between concepts using the relationship in the knowledge structure. Self-attention-based KT models such as Self-Attentive Knowledge Tracing (SAKT) (Pandey & Karypis, 2019), Attentive Knowledge Tracing (AKT) (Ghosh et al., 2020), and Hierarchical Attention Knowledge Tracing (HAKT) (Sun, Zhou, Liu, He, & Tang, 2021). AKT couples attention-based neural network models with a series of interpretable model components to model concept mastery. At the question level, it uses the Rasch model to capture individual differences between questions.

While models like LPKT (Shen et al., 2021) and HawkesKT (Wang et al., 2021) have their own merits in focusing on the student learning process, they tackle the problem from a different angle. LPKT is designed to align the changes in a student's knowledge state with the learning process, capturing variations between successive learning units and considering the impact of a forgetting factor. As a result, it achieves a reasonable representation of knowledge state and performs well. Similarly, HawkesKT makes a notable contribution by empirically capturing temporal cross-effects in KT through the innovative use of the Hawkes process, demonstrating advantages in both performance and training efficiency. However, what sets our work apart is its distinct approach. Rather than focusing primarily on the temporal aspects, our work initiates from the perspective of student cognition. We methodically break down the steps a student takes while working on each exercise, offering a multifaceted causal analysis of the response outcomes. Consequently, our model excels in providing interpretable predictive results, a feature that distinguishes it from existing models.

2.3. IRT-based knowledge tracing

The Item Response Theory (IRT) provides a statistical model for the probability of a student answering a question correctly. Its history can be traced back to Thurston's pioneering work in the 1920s (Tatsuoka, Lord, Novick, & Birnbaum, 1971) and subsequent research (Baker & Kim, 2004; Green, 1951; Lord, 1951, 1980). The core idea is to estimate student performance through a logistic function, taking into account both subjective factors like concept mastery (Masters, 1982) and objective factors like item difficulty (Hambleton & Swaminathan, 2013), item discrimination (Deonovic, Yudelson, Bolsinova, Attali, & Maris, 2018), guessing rate (Birnbaum, 1968), slip rate (Barton & Lord, 1981), and so on. The Rasch model (Rasch, 1993), a form of IRT, uses a single-parameter logistic regression model to define the probability of a correct answer. The two-parameter logistic model (2PLM) extends this by adding an item discrimination parameter.

While traditional IRT models provide psychologically interpretable parameters, they may not capture all the nuances in the data. In contrast, neural networks offer better data exploitation but lack interpretability. Combining these two approaches aims to leverage their respective strengths.

In 2014, Mohammad et al. proposed a fusion of the IRT and KT paradigms (Khajah et al., 2014) to achieve good evaluation results and student learning modeling capabilities. They combined the Rasch model in Item Response Theory and the Hidden Markov Models in knowledge tracing. In 2019, Chun-Kit proposed the Deep-IRT model (Yeung, 2019) based on DKVMN, which integrates the IRT model to estimate student ability and item difficulty, improving the model's interpretability and exploring the correlation between estimated item difficulty and other sources of item difficulty. In 2021, Emiko Tsutsumi et al. proposed the Deep-IRTw model (Tsutsumi et al., 2021), which also integrates the IRT model into DKVMN. Unlike Deep-IRT, they construct two independent neural networks: a student network and an item network. They divide item difficulty into concept difficulty and item difficulty in the item network, and use the sum of the two difficulties as the item difficulty in the one-parameter logistic model(1PLM). This approach improves predictive accuracy while also enhancing interpretability compared to Deep-IRT.

In Section 4.2.2, we leverage the Two-Parameter Logistic Model (2PLM) to assess question difficulty, which enables us to gauge a student's mastery over a specific question before accounting for random guessing behaviors. Our model outperforms Deep-IRT in two key areas: predictive accuracy and interpretability. Notably, we generate a more nuanced curve that accurately charts the shifts in levels of concept mastery, and we offer more precise estimations of question difficulty.

3. Problem definition

The objective of Knowledge Tracing (KT) is to dynamically monitor a student's evolving knowledge state, using their historical data, and to make informed predictions about their future responses to questions. For a given student, we represent the historical record of their answers to T questions as $\boldsymbol{X} = [\boldsymbol{x}^{\langle 1\rangle}, \dots, \boldsymbol{x}^{\langle t\rangle}, \dots, \boldsymbol{x}^{\langle T\rangle}]$. In this notation, $\boldsymbol{x}^{\langle t\rangle}$ denotes the record at time step t, generally comprising a triplet $(c^{\langle t\rangle}, q^{\langle t\rangle}, a^{\langle t\rangle})$. Here, $c^{\langle t\rangle} \in \mathbb{N}^+$ indicates the concept index at time step t, $q^{\langle t\rangle} \in \mathbb{N}^+$ is the question index, and $a^{\langle t\rangle} \in \{0, 1\}$ symbolizes the student's answer, where 1 signifies a correct response and 0 an incorrect one. As the focus here is on individual student predictions, we exclude the subscript i which would otherwise denote different students.

For our phase-structured modeling of the learning process, the task is articulated as follows: Given each student's historical answering pattern $\boldsymbol{X} = [(c^{\langle 1\rangle}, q^{\langle 1\rangle}, a^{\langle 1\rangle}), \dots, (c^{\langle t\rangle}, q^{\langle t\rangle}, a^{\langle t\rangle}), \dots, (c^{\langle T\rangle}, q^{\langle T\rangle}, a^{\langle T\rangle})]$, we aim to: (1) track fluctuations in their knowledge states and estimate their mastery levels across all concepts at various time steps T; (2) gauge the difficulty level β of each question they have responded to; (3) ascertain their probabilities of guessing and slipping, denoted $P(G)$ and $P(S)$, for each attempted question; and (4) predict their forthcoming response $a^{\langle T+1\rangle}$ to the question $q^{\langle T+1\rangle}$ related to the concept $c^{\langle T+1\rangle}$ at time step $T + 1$.

4. Method

For a student, PKT models the student's learning process stage by stage.

Modeling mastery of concepts: starting from the abstract and shallow thinking stage, the model captures the student's mastery of a concept. PKT follows the idea of active construction and insight from constructivist learning theory to construct the loss function for the degree of mastery of a concept.

Modeling solving questions: at the stage of deep thinking, modeling the degree of solving questions. The interaction between a student's conceptual mastery and the characteristics of the question will form the student's degree of solving a question. Inspired by 2PLM, the student's mastery of a concept, the difficulty and the discrimination of the question are constrained together to characterize the student's ability to solve a question.

Modeling answering behaviors: finally, the model considers the possibility of random guessing or slipping by the student when answering questions, which may obscure the student's true understanding of concepts and questions. At this stage, PKT integrates the solving of the question and the student's answering behavior to predict whether the student can answer the next question correctly.

PKT is divided into two modules: the encoding module and the progressive learning process module. The encoding module is used to embed and encode the input information, while the progressive learning process module simulates the student's learning process. In practice, the encoding module uses the powerful modeling capabilities of deep learning to capture latent features from the student's historical question-concept-answer interactions and decode parameters that have educational significance. The progressive learning process module unfolds the estimation of the learning process in stages in a sequence from the inside out and from abstract to concrete.

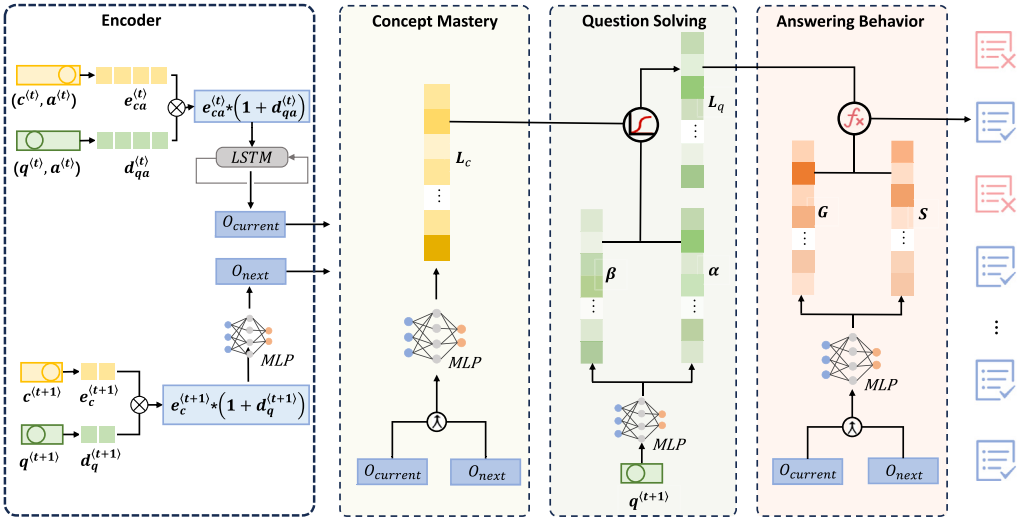

Fig. 2. Overview of the PKT method.

4.1. Encoding module

We use four encoders in our model. For the original feedback sequence, we embed "concept-answer" and "question-answer" separately. For the information at time $t+1$, we embed "concept" and "question" separately.

First, we embed the information at time t. The concept $c^{\langle t\rangle}$ and the student's response $a^{\langle t\rangle}$ are embedded at the concept level, resulting in an output of $\boldsymbol{e}_{ca}^{\langle t\rangle} \in \mathbb{R}^N$, which represents the embedding of the concept and the answer. The question $q^{\langle t\rangle}$ and the student's response $a^{\langle t\rangle}$ are embedded at the question level, resulting in an output of $\boldsymbol{d}_{qa}^{\langle t\rangle} \in (0,1)^N$, which represents the embedding of the question and the answer. Here, N is the size of embedding vector.

We first combine two embedding vectors, $\boldsymbol{e}_{ca}^{\langle t\rangle}$ and $\boldsymbol{d}_{qa}^{\langle t\rangle}$, into $\boldsymbol{e}_{ca}^{\langle t\rangle} * (1+\boldsymbol{d}_{qa}^{\langle t\rangle})$. This joint encoding effectively captures the intricate relationships between concepts and questions. The rationale behind this is that while the concept, represented by $\boldsymbol{e}_{ca}^{\langle t\rangle}$, provides coarser-grained information, the question supplies finer-grained information that supplements the concept, denoted by $(1+\boldsymbol{d}_{qa}^{\langle t\rangle})$. Both $\boldsymbol{e}_{ca}^{\langle t\rangle}$ and $\boldsymbol{d}_{qa}^{\langle t\rangle}$ are sigmoid-activated, thus lying in the range $(0,1)$. Consequently, $(1+\boldsymbol{d}_{qa}^{\langle t\rangle})$ serves as a scaling factor in the range $(1,2)$ to adjust and amplify the conceptual encoding $\boldsymbol{e}_{ca}^{\langle t\rangle}$.

Next, for the time step $t+1$, the concept $c^{\langle t+1\rangle}$ is embedded as $\boldsymbol{e}_{c}^{\langle t+1\rangle} \in \mathbb{R}^N$, and the question $q^{\langle t+1\rangle}$ as $\boldsymbol{d}_{q}^{\langle t+1\rangle} \in (0,1)^N$. In a manner akin to the previous step, these two embeddings are integrated to form $\boldsymbol{o}_{next} = \boldsymbol{e}_{c}^{\langle t+1\rangle} * (1+\boldsymbol{d}_{q}^{\langle t+1\rangle})$.

As illustrated in Fig. 2, the Encoder module is partitioned into two functional segments. The upper segment focuses on encoding the historical data from time steps 1 through t, which inherently carries sequential attributes. To adeptly manage this temporal information, an LSTM (Long Short-Term Memory) model is utilized, generating a dynamic student state encapsulated as $\boldsymbol{o}_{current}$. Conversely, the lower segment is responsible for encoding the information about both the concept and the question at the subsequent time step $t+1$. This is achieved using a Multi-Layer Perceptron (MLP), yielding a static embedding for the question and the concept at that particular time, denoted as $\boldsymbol{o}_{next}$.

We concatenate $\boldsymbol{o}_{current}$ and $\boldsymbol{o}_{next}$ to obtain $\boldsymbol{o} = \boldsymbol{o}_{current} \oplus \boldsymbol{o}_{next}$, and then send $\boldsymbol{o}$ to the decoder for decoding, resulting in:

$$\boldsymbol{P} = \sigma\left[MLP_P(\boldsymbol{o})\right], \boldsymbol{P} \in \left\{\boldsymbol{L_c}, \boldsymbol{S}, \boldsymbol{G}\right\} \tag{1}$$

where $\sigma(\cdot)$ is the sigmoid function. The vector $\boldsymbol{o}$ is processed by a multilayer perceptron denoted as $MLP_P(\cdot)$, which generates the vector parameters $\boldsymbol{P}$. The vector parameters $\boldsymbol{P}$ can take the values of the parameters in the curly braces, including the concept-related parameter $\boldsymbol{L_c}$, as well as the answering behavior-related parameters $\boldsymbol{S}$ and $\boldsymbol{G}$. The latter two parameters will be further explained below.

Next, we introduce the progressive learning process module.

4.2. Progressive learning process module

According to the hypothesis of progressive learning process modeling, the learning process module builds models for three stages: concept mastery, question solving, and answering behavior. This module follows certain educational theories to design loss functions that incorporate interpretable parameters to adapt to different stages of the learning process.

4.2.1. Concept mastery

The concept mastery part models students' understanding and mastery of concepts. Constructivist learning theory (Bruner et al., 1966; Cobb & Bowers, 1999; Duffy & Jonassen, 2013) posits that learning is an active process in which learners use their existing knowledge and experiences as a starting point to actively construct new knowledge and experiences. Therefore, PKT continuously updates a student's mastery level of all concepts based on the time step, building on their existing level of mastery. In the concept mastery part of PKT, the output of the encoding module is decoded to construct the concept mastery level $\boldsymbol{L_c}$ of each student at each time step. Specifically, $\boldsymbol{L_c} = [L_{c_1}, L_{c_2}, \ldots, L_{c_j}, \ldots, L_{c_{N_c}}]$, where N_c represents the number of concepts, $L_{c_j} \in [0,1]$ denotes a student's mastery level of the jth concept, which changes over time.

We design a tiered penalty mechanism for the loss function of the concept mastery parameter. The model will not be penalized when the change in concept mastery is within a reasonable range, but will be penalized for the part that exceeds the reasonable range. The loss of $\boldsymbol{L_c}$ on the jth concept $loss_{L_{c_j}}$ is:

$$loss_{L_{c_j}} = \begin{cases} \Delta L_{c_j}^2, & \Delta L_{c_j} \in (-\infty, \Gamma_{lower}) \\ 0, & \Delta L_{c_j} \in [\Gamma_{lower}, \Gamma_{upper}] \\ \Delta L_{c_j}^2, & \Delta L_{c_j} \in (\Gamma_{upper}, +\infty) \end{cases} \tag{2}$$

In Eq. (2), the term ΔL_{c_j} quantifies the shift in a student's proficiency regarding the jth concept as the time step progresses from t to $t+1$. Explicitly, ΔL_{c_j} is defined as the difference $L_{c_j}^{\langle t+1\rangle} - L_{c_j}^{\langle t\rangle}$. When ΔL_{c_j} changes by more than Γ_{upper} or decreases by more than Γ_{lower} between two consecutive time steps, the model will be penalized with $\Delta L_{c_j}^2$.

The active construction and possible "insight" phenomenon (Dewey, 1910; Perkins, 1981) in a student's learning process indicate that

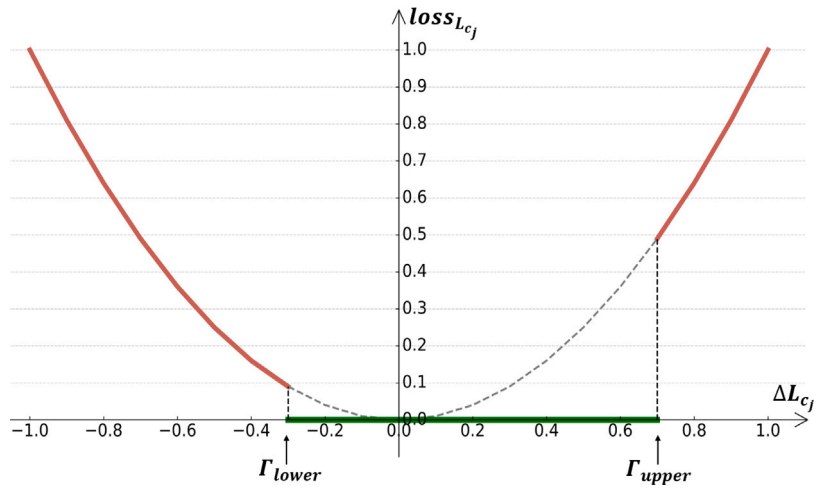

Fig. 3. A hierarchical penalty mechanism is designed for the level of mastery parameter L_c of a concept.

his/her concept mastery tends to increase as the student practices more and may suddenly spike. Therefore, the tiered penalty mechanism is designed to have a lenient penalty for progress and a strict penalty for regressions, i.e., the upper limit for penalties is greater than the absolute value of the lower limit.

As shown in Fig. 3, the red area represents the part that is penalized for exceeding the reasonable range of concept mastery change, while the green area represents the part that is within the reasonable range and will not be penalized. In practice, we set $\Gamma_{upper} = 0.7$ and $\Gamma_{lower} = 0.3$. The range setting produces a smooth change curve for the concept mastery state $\boldsymbol{L_c}$, gradually increasing as the student masters the question. This curve has directionality and readability, and rarely exhibits wave-like transitions that can mislead students' understanding of their knowledge concept state (Yeung & Yeung, 2018). At the same time, it also performs well in terms of performance results. The total loss of $\boldsymbol{L_c}$ is as below:

$$loss_{L_c} = \sum_{j=1}^{N_c} loss_{L_{c_j}} \tag{3}$$

4.2.2. Question solving

Whether students can answer a question correctly depends not only on their own concept mastery level, but also on the their understanding of the question. The characteristics of the question itself, such as its difficulty and discrimination, also affect students' solving of the question. We use MLP to fit the features of the question:

$$\boldsymbol{Q} = \sigma\left[MLP_Q\left(q^{\langle t+1\rangle}\right)\right], \boldsymbol{Q} \in \{\boldsymbol{\alpha}, \boldsymbol{\beta}\} \tag{4}$$

where $MLP_Q(\cdot)$ represents a multilayer perceptron used to process $q^{\langle t+1\rangle}$ and generate the question-related vector parameters $\boldsymbol{Q}$. The vector parameters $\boldsymbol{Q}$ take the parameters from curly braces. $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_j, \ldots, \alpha_{N_c}]$, where $\alpha_j \in [0, 1]$ represents the discrimination of a particular question under concept j. $\boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_j, \ldots, \beta_{N_c}]$, where $\beta_j \in [0, 1]$ represents the difficulty of a question under concept j.

In this module, in order to simulate the deep interaction with concepts and the question in students' minds, the model integrates the degree of concept mastery and the characteristics of the question to obtain their solving level $\boldsymbol{L_q}$ for a particular question. Inspired by the classic 2PLM (two-parameter logistic model) in IRT, we use the response function shown as:

$$L_{q_j} = \frac{e^{4\alpha_j\left(L_{c_j}-\beta_j\right)}}{1 + e^{4\alpha_j\left(L_{c_j}-\beta_j\right)}} \tag{5}$$

By using students' concept mastery L_{c_j} for concept jth at time t, along with the corresponding difficulty β_j and discrimination α_j of a particular question under concept jth, we can predict students' mastery level for that question. In the classic 2PLM, the constant typically multiplied by the discrimination parameter α is 1.7. However, in order to fully utilize the nonlinearity of the sigmoid function, we amplify it to 4 (Yeung, 2019). After applying Eq (5), taking into account the joint effects of concept mastery and question solving, we obtain the probability $\boldsymbol{L_q}$ of a student correctly answering a question without considering random answering behavior.

4.2.3. Answering behavior

After determining the question-solving strategy, students start to write down the answer. Drawing on the consideration of answer behaviors in the classic BKT model (Corbett & Anderson, 1994), PKT divides the behaviors that are inconsistent with the mastery of concepts into two types when students are answering questions. One is when students guess and answer a question correctly without really solving the question (corresponding to the first term in Eq (6)); the other is when students answer a question incorrectly despite being capable of solving it (corresponding to the second term in Eq (6)). Summing the probabilities of the two conditions, the probability that a student will get a question right is:

$$\hat{a}^{\langle t+1\rangle} = \boldsymbol{L_q} * (1 - \boldsymbol{S}) + \left(1 - \boldsymbol{L_q}\right) * \boldsymbol{G} \tag{6}$$

The answering behavior module decodes the guessing parameter $\boldsymbol{G}$ and slipping parameter $\boldsymbol{S}$ from the historical feedback sequence and designs a gate threshold loss function for optimization. Regarding the loss of $\boldsymbol{G}$ and $\boldsymbol{S}$, we adopt a tiered penalty mechanism. When the value of guessing or slipping is within a reasonable range, no penalty is imposed, otherwise a penalty is imposed. The loss of $\boldsymbol{S}$ on the jth concept is denoted as $loss_{S_j}$ and is calculated as follows:

$$loss_{S_j} = \begin{cases} 0, & \Delta S_j \in (-\infty, 0] \\ \Delta S_j, & \Delta S_j \in (0, +\infty) \end{cases} \tag{7}$$

Here, $\Delta S_j = \sum_{\tau=1}^{t+1} S_j^{\langle\tau\rangle} - \lambda_s * len_j$. $S_j^{\langle\tau\rangle}$ represents the slipping probability of a student on the jth concept at the τth time step. $\sum_{\tau=1}^{t+1} S_j^{\langle\tau\rangle}$ is the sum of all the slipping probabilities of a student on the jth concept for all questions generated by the model. $\lambda_S \in [0, 1]$ is a manually set upper limit for the slipping probability of each question (in PKT, we set $\lambda_S = 0.4$). $len_j \in \mathbb{N}^+$ is the number of questions on the jth concept. $\lambda_s * len_j$ is the maximal slipping probability of all questions on the jth concept for reference.

Finally, we obtain the total loss of $\boldsymbol{S}$ by summing up the losses on each concept, where N_c indicates the number of concepts:

$$loss_{\boldsymbol{S}} = \sum_{j=1}^{N_c} loss_{S_j} \tag{8}$$

The guessing parameter $\boldsymbol{G}$, which represents the probability of a student guessing, has a similar loss function. We also set the manually set upper limit for the guessing probability of each question to $\lambda_G = 0.4$.

Finally, the predicted loss and parameter loss are combined into the total loss:

$$loss = loss_{L_c} + loss_S + loss_G + loss_{Pred}, \tag{9}$$

Here, $loss_{Pred}$ is the binary cross entropy of prediction and target.

5. Experiments

We will introduce the experimental setup in Section 5.1, discuss the experimental results in Section 5.2, explain the learning process in Section 5.3, and demonstrate reliable question difficulty estimation in Section 5.4.

5.1. Datasets and evaluation metrics

Experiments are conducted on six commonly used publicly available datasets in the field of knowledge tracing, including Assist09, Assist17, Static11, Eedi, EdNet, and FSAI-F1toF3. The ASSIST dataset series are provided by the ASSISTments online intelligent tutoring system. **Assist09** (Feng, Heffernan, & Koedinger, 2009) and **Assist17** datasets are consistent with the datasets used in AKT model. **Static11** (Koedinger et al., 2010) is gathered from an engineering statics course at a university. **Eedi** (Wang et al., 2020) was used in the NeurIPS 2020 Education Data Mining Challenge[1] and was collected by the online education platform Eedi from 2018 to 2020. We use the data provided in the third task of the competition, which includes the stem information of 948 questions. **EdNet** is a dataset collected by the online tutoring platform Santa from 2017 to 2019. **FSAI-F1toF3** (Yeung, 2019) is provided by the Find Solution Ai Limited and is collected via an adaptive learning tablet application called 4LittleTrees.[2] In the Deep-IRT paper, they extracted the student interactions that are related to mathematics curriculum from F.1 to F.3 in Hong Kong.[3] For this dataset, our data is consistent with that used in the Deep-IRT model (Yeung, 2019).

In this experiment, we preprocess the datasets as follows: For the Assist09 dataset, we use the corrected dataset to address issues with data duplication (Xiong, Zhao, Van Inwegen, & Beck, 2016). For the Static11 dataset, we combine the original question and step numbers to create a new question ID as input, and we only keep the first record of multiple consecutive answers to the same question. When loading the data, we remove answer sequences with a length less than 3, and for efficiency purposes, we split answer sequences longer than 200 into multiple sequences with a length between 3 and 200 (Ghosh et al., 2020). For the Eedi dataset, we randomly select the first 200,000 answer records for experimentation, involving 4,918 students and 930 questions. For the EdNet dataset, due to its large size, we randomly select data from 5,000 students for experimentation (Yang et al., 2021). To ensure the reasonability of answer sequence length and for efficiency purposes, we remove answer sequences longer than 1,000 or shorter than 10, resulting in data from 4,611 students after preprocessing. The statistical information of the preprocessed datasets is shown in Table 2.

In this experiment, we apply 5-fold cross-validation to all models and datasets. For each fold, 60% of the sample data is used for training, 20% for validation, and 20% for testing. For each fold, we train the models with the training set, perform early stopping to find the best parameters for each model on the validation set, and test the trained models on the test set. We record the average of the model's five runs on the test set as the final result and calculate the standard deviation to reflect the variability of the test results. We record the results in the form of $mean \pm std$ as shown in Table 3.

Table 2
Statistics of the dataset after preprocessing.

Dataset	Students	Concepts	Questions	Responses
Assist09	4,151	110	16,891	325,637
Assist17	1,709	102	3,162	942,816
Static11	331	106	633	112,921
Eedi	4,918	53	930	200,000
EdNet	4,611	1,670	11,103	523,779
FSAI-F1toF3	310	99	2,266	51,283

To report the performance of the models, we use two metrics: (1) We employ Area Under Curve (AUC) as the criterion for models comparison. AUC provides a robust measure for binary prediction evaluation. When interpreting AUC values, a higher AUC indicates that the model predicts more accurately. AUC = 0.5 indicates that the prediction is equal to random guessing. (2) We use the accuracy (ACC) to represent the consistency between the predicted and true scores of the students. If the predicted probability of answering the next question correctly is 0.5 or higher, the student is predicted to answer correctly; otherwise, it is incorrect.

5.2. Comparative study

This section compares our models with several benchmark models, including DKT (Piech et al., 2015), DKVMN (Zhang et al., 2017), AKT (Ghosh et al., 2020), Deep-IRT (Yeung, 2019), EnKT$_b$ (Sun et al., 2022), and LPKT (Shen et al., 2021).

Among them, the classical models DKT and DKVMN cannot simultaneously incorporate both concept and question inputs. The AKT model takes both question and concept IDs as inputs, and uses the Rasch model to construct raw question and concept embeddings. The Deep-IRT model combines IRT and DKVMN based on deep neural networks, and it takes questions or concepts as inputs. In Deep-IRT paper, the model uses concept tags as input on the Assist09 dataset, and question tags as input on the Static11 and FSAI-F1toF3 datasets (Yeung, 2019). This experiment follows the same approach. The EnKT$_b$ model is the basic part of the Ensemble Knowledge Tracing (EnKT) model without the ensemble method. Based on the hybrid of BKT and DKT, EnKT$_b$ models the pairwise interactions between the three key objects in the learning process: students, concepts, and questions. Compared with AKT, its basic model has already demonstrated superior performance. Therefore, EnKT$_b$ is a SOTA KT model (Sun et al., 2022). The LPKT model[4] (Shen et al., 2021) captures learning gains and considers the forgetting phenomenon during the learning process. Given that none of the models compared in this experiment utilize time information, for fairness, the model compared is the LPKT(no time), which drops both answer time and interval time.

The comparative experimental results are shown in Table 3.

Overall, PKT's performance is comparable to EnKT$_b$, and outperforms both AKT and LPKT(no time) in 5 out of the 6 datasets. Compared to AKT, PKT outperforms it by 4% on the FSAI-F1toF3 dataset and shows improvements of 0.6% to 1.2% on the Assist17, Eedi, and EdNet datasets. On the Static11 dataset, AKT performs slightly better than PKT by 0.25%. We speculate that this is due to the small number of questions in the dataset, which only has 633 preprocessed questions. In PKT, modeling the questions is a key element in modeling the entire learning process, and having too few questions may limit PKT's performance. Compared with EnKT$_b$, we surpass it on most datasets, although we fall slightly behind on a few datasets. However, explanatory experiments show that this is due to EnKT$_b$ sacrificing some interpretability (refer to

[1] https://eedi.com/projects/neurips-education-challenge
[2] https://www.4littletrees.com/.
[3] https://github.com/ckyeungac/DeepIRT
[4] https://github.com/bigdata-ustc/EduKTM

Table 3
Prediction results of AUC and ACC. We run each model five times and reported the averaged results and standard deviations in the form of *mean* ± *std*. The best performers are shown in boldface and the second best in italics.

		DKT	DKVMN	AKT	Deep-IRT	EnKT$_b$	LPKT(no time)	PKT
Assist09	AUC	80.65 ± 0.40	79.72 ± 0.35	*82.90 ± 0.33*	81.54 ± 0.10	82.88 ± 0.33	75.16 ± 0.75	**82.99 ± 0.27**
	ACC	76.57 ± 0.64	75.84 ± 0.62	77.62 ± 0.54	76.86 ± 0.07	*77.70 ± 0.53*	71.11 ± 1.12	**77.72 ± 0.41**
Assist17	AUC	68.86 ± 0.21	68.19 ± 0.29	75.59 ± 0.29	73.06 ± 0.06	76.05 ± 0.29	**78.95 ± 0.52**	*76.23 ± 0.30*
	ACC	67.67 ± 0.59	67.21 ± 0.56	70.74 ± 0.38	69.50 ± 0.08	71.01 ± 0.34	**73.12 ± 0.48**	*71.04 ± 0.37*
Static11	AUC	81.34 ± 0.50	84.32 ± 0.64	**86.59 ± 0.50**	82.78 ± 0.37	*86.37 ± 0.28*	85.87 ± 0.28	86.34 ± 0.32
	ACC	79.97 ± 0.68	80.45 ± 0.70	81.96 ± 0.42	81.09 ± 0.17	*82.44 ± 0.36*	82.24 ± 0.40	**82.46 ± 0.34**
Eedi	AUC	71.10 ± 0.39	70.33 ± 0.38	74.75 ± 0.36	74.55 ± 0.07	*75.54 ± 0.34*	73.26 ± 0.58	**75.55 ± 0.31**
	ACC	65.53 ± 0.30	64.91 ± 0.31	68.36 ± 0.33	68.40 ± 0.11	**69.08 ± 0.46**	67.80 ± 0.47	*68.91 ± 0.36*
EdNet	AUC	71.86 ± 0.32	69.96 ± 0.35	73.24 ± 0.33	74.06 ± 0.06	*74.43 ± 0.37*	69.26 ± 0.77	**74.45 ± 0.41**
	ACC	68.44 ± 0.37	67.69 ± 0.41	69.35 ± 0.24	69.57 ± 0.17	*69.88 ± 0.35*	66.29 ± 0.50	**69.92 ± 0.34**
FSAI-F1toF3	AUC	60.50 ± 0.38	60.31 ± 0.64	64.42 ± 1.31	67.08 ± 0.05	**68.48 ± 0.65**	67.21 ± 0.95	*68.13 ± 0.66*
	ACC	58.52 ± 0.43	58.21 ± 0.72	60.76 ± 1.02	61.71 ± 0.18	*62.86 ± 0.50*	**63.70 ± 0.75**	62.53 ± 0.62

Section 5.3). Psychometric model IRT can enhance interpretability, and among models that combine IRT (AKT, Deep-IRT, PKT), PKT performs the best. Compared with Deep-IRT, PKT has higher AUC and ACC on all datasets. Besides performance, in the subsequent explanatory experiments, PKT demonstrates more reasonable interpretability in visualizing the learning process and estimating question difficulty closer to real-world difficulty. Compared with LPKT(no time), PKT excels on five datasets, especially on Assist09 and EdNet with an AUC increase of over 5%. On Assist17, while PKT does not surpass LPKT(no time), it still outperforms other models. It is important to note that while PKT may not lead on certain datasets compared to SOTA models (AKT, EnKT$_b$, LPKT(no time)), it stands out in terms of interpretability. Sections 5.3 and 5.4 will showcase experiments highlighting PKT's strengths in interpretability.

We attribute PKT's good performance to its more realistic modeling process. Firstly, PKT's superiority over DKT and DKVMN demonstrates the effectiveness of modeling both question-solving and answering. Even if students have already mastered a concept, they may still struggle with question difficulty during the solving process, as well as random errors during the answering process, leading to incorrect answers. Therefore, intuition and experiments have shown that only considering concept mastery is not comprehensive for predicting a student's future performance. Secondly, PKT's superiority over AKT and Deep-IRT demonstrates that small probability events (guessing and slipping) that are not consistent with concept mastery cannot be ignored in the answering process. Thirdly, PKT has slightly better performance than EnKT$_b$, but its interpretability is much better. PKT reduces the impact of random guessing behavior. EnKT$_b$ sets the upper limit of punishment for guessing at 0.8. Such lenient penalty limits result in too many correct answers being explained by guessing, ensuring performance but sacrificing interpretability. In contrast, PKT sets a more stringent upper limit of punishment at 0.4, providing a more reasonable explanation while maintaining slightly better performance. Additionally, PKT uses the 2PLM model to fit a more reasonable learning process and more accurate question difficulty, partially compensating for the performance loss caused by the stricter upper limit of punishment and improving the model's interpretability. On the majority of datasets, PKT outperforms LPKT(no time) in performance. We attribute this advantage to PKT's more granular, stage-wise modeling approach of the learning process.

5.3. Explain the learning process

According to the sequence from the abstract to the concrete modeling, PKT models the concept mastery, question-solving, and answering behaviors. In the concept mastery stage, the abstract concept mastery degree L_c is modeled. Gradually transitioning to the specific question-solving and answering behaviors stage, PKT models various factors, such as item difficulty β and guess probability $P(G)$. These factors are captured explicitly in these later stages, aiding us in assessing the degree to which they influence a student's ability to apply abstract understanding to tangible problems.

Analyzing the modeling process of PKT can help us understand why PKT provides such predictions, and also help students estimate areas where they may have gaps in understanding. Provide sufficient explanations for binary answer results (true or false) to facilitate the use of both Knowledge of Results and Knowledge of Performance feedback to help improve students' learning effectiveness.

To analyze the decision-making process of PKT, we conduct a learning process feedback analysis experiment. As illustrated in Fig. 4, we randomly select four students from the Assist09 dataset and visualize their individual knowledge concept learning processes.

We find that the learning process graph fitted by PKT is more consistent with educational intuition. Firstly, PKT's mastery curve of knowledge level is more stable, avoiding the misleading wave-like transitions that can easily confuse students. Secondly, PKT can analyze the causes of answer results and provide more informative explanations, resulting in a more reasonable explanation process. Thirdly, for small probability events during the learning process (guessing or slipping), PKT's descriptions are more accurate. Fourthly, PKT models the difficulty of questions more reasonably (see Section 5.4 for details). Specific examples will be provided below for further illustration.

(1) PKT provides a more stable fit for the knowledge mastery curve. From an educational perspective, students' knowledge concept states should gradually change directionally over time, rather than alternating dramatically between mastered and unmastered states. Wave-like transitions can be misleading for students' understanding of their knowledge states (Yeung & Yeung, 2018).

As seen in the first three examples, the Deep-IRT concept mastery curve is irregular with sharp drops and sudden increases in mastery level. In comparison, the EnKT$_b$ fitting curve is relatively stable. Among the three models, the student concept mastery curve fitted by PKT has the least amount of fluctuation, with fewer occurrences of sudden drops, displaying the trend of learners' concept mastery level changes more reasonably and directionally. This indicates that the design of a more relaxed progress penalty and a more strict backdown penalty in the grading punishment mechanism for the concept mastery state $\boldsymbol{L_c}$ is reasonable.

(2) PKT provides richer information to explain the learning process. While fitting the concept change curve, PKT generates reasonable parameter explanations for students' performance, providing the current question's difficulty (D), discrimination (α), guessing probability (G), and slipping probability (S) for each feedback.

Compared to Deep-IRT, which typically explains a student's wrong answers through sudden drops in concept mastery, PKT can attribute wrong answers to high question difficulty or discrimination, or potential slipping in the student's response behavior. Consider the case of $id_{stud} = 46$ and $id_{conc} = 44$, referred to as (46,44) hereinafter. After

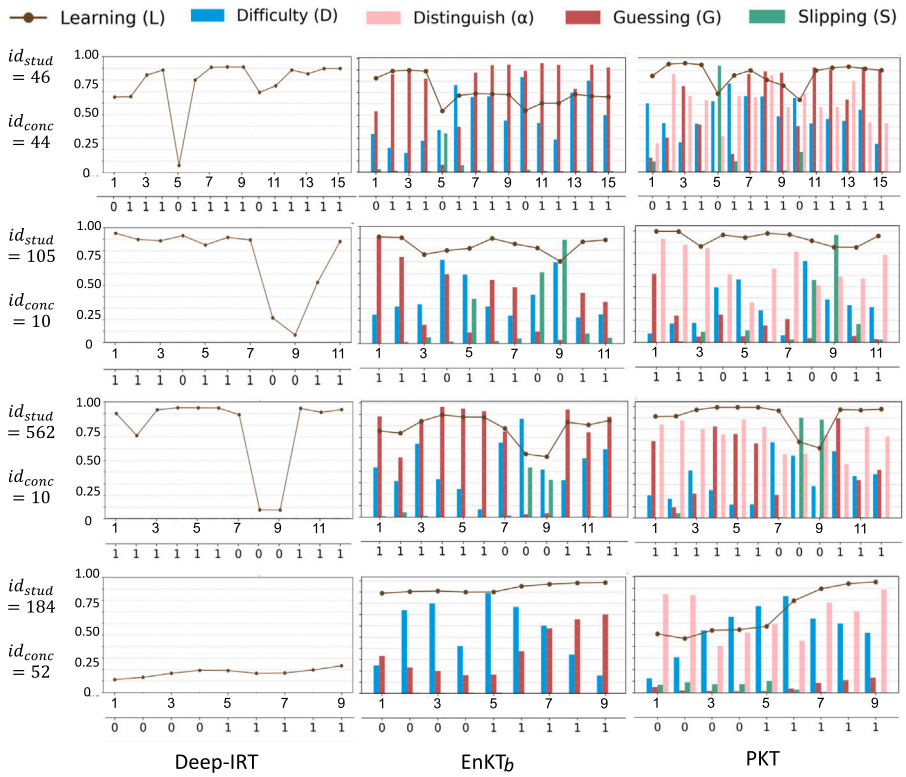

Fig. 4. Comparing the interpretability of Deep-IRT, EnKT$_b$, and PKT for specific learning processes. Here, id_{stud} and id_{conc} represent the ID of the student and concept, respectively.

answering three consecutive questions correctly, the student answered the fifth question incorrectly. Deep-IRT explains this as a sharp drop in the student's mastery level from 0.9 to 0.1. In contrast, PKT attributes the incorrect answer primarily to slipping, as indicated by the length of the green stripe in the graph, and secondarily to a moderately difficult question, as indicated by the blue stripe above 0.6, and the student's mastery level only slightly decreased.

The changes in mastery level provided by PKT are more realistic and the explanations for errors are more reasonable. In addition, the sudden drop in Deep-IRT in examples (105,10) and (562,10) also has a more reasonable fitting process and causal explanation in PKT.

Compared to Deep-IRT, PKT can use rich information from questions and response behavior to analyze why learners make errors on specific test questions, demonstrating greater superiority and reliability in simulating the learning process. The comprehensive and reasonable process knowledge feedback provided by PKT can also help students improve their learning efficiency.

Compared to EnKT$_b$, PKT also demonstrates advantages in providing more reasonable explanations for the learning process. For example, in the last case, as the student gradually masters a concept, PKT's analysis of the student's answers to the last five questions shows that when the difficulty of the questions exceeds 0.5 and the discrimination is also high, the student is able to answer correctly because the student's mastery level has improved. In contrast, EnKT$_b$ assumes little change in the student's concept mastery level throughout the learning process and attributes five consecutive correct answers to easy questions and high guessing probabilities. However, in cases where the student answers incorrectly at the beginning and consecutively answers correctly, PKT's gradual improvement in mastery level explanation is more reasonable.

(3) PKT reduces the impact of random guessing. Our intuition is that as students gradually master a concept, the likelihood of correctly answering a question due to their knowledge concept mastery should increase, while the contribution of blind guessing should decrease. Random occurrences such as guessing and slipping should not be the main factors in explaining the answer results.

EnKT$_b$ has a high overall guessing probability, as shown in the prominent red striped lengths in the figure, which even exceed the level of mastery of a concept in most cases. Sometimes, even when answering a question incorrectly, the probability of guessing correctly is still high, which is not realistic. For instance, in the EnKT$_b$ model, question 10 of (46,44), question 4 of (105,10), and question 7 of (562,10) all exhibit high guessing probabilities. By contrast, PKT punishes guessing parameters more strictly in the loss function, which reduces the excessive influence of the random guessing factor G. As a result, PKT attributes more correct answers to the improvement of students' mastery level rather than guessing.

The above experimental analyses indicate that PKT performs well in explaining the learning process, and its simulated learning process graph is more in line with educational reality. Additionally, PKT can also account for some confusing phenomena in Deep-IRT and EnKT$_b$. In the next experiment, we will verify the reliability and rationality of the question difficulty learned by PKT, which is closer to real-world difficulty levels.

5.4. Study the difficulty of the questions

This section evaluates the difficulty of questions from two perspectives and explores the distribution of question difficulty in the dataset used.

5.4.1. Assess difficulty reliability and rationality

(1) From a statistical perspective: Assessing the correlation of question difficulty

To evaluate the difficulty levels estimated by PKT, we compared the difficulty levels learned by PKT from the FSAI-F1toF3 dataset with those obtained from two other sources. We use the same dataset, difficulty sources, and Pearson correlation evaluation method as in Deep-IRT to ensure a fair comparison.

We select five concepts that accounted for approximately one-fifth of the interactions in the FSAI-F1toF3 dataset, totaling 131 questions. These concepts are "Significant Figures", "Approximation and

Table 4
Pearson correlation test. Comparing the correlation between Deep-IRT and PKT on the difficulty level from two sources, publisher difficulty and item analysis difficulty. The ones with stronger correlations are represented in bold.

	Publisher difficulty	Item analysis difficulty
Deep-IRT	0.08	0.56
PKT	**0.24**	**0.78**

Errors in Measurement", "Index Notation", "Laws of Indices", and "Polynomials".

Since students could answer the same question multiple times in this dataset, we only use their first attempts when calculating difficulty levels and considered questions that had been answered by at least 10 different students. This resulted in a total of 109 questions analyzed, consistent with the selection method used in the Deep-IRT (Yeung, 2019). The following are the three sources of difficulty:

(1) The first source of difficulty level is provided by the publisher. The FSAI-F1toF3 dataset provides a difficulty level $\in \{1, 2, 3\}$ for each question, representing easy, medium, and hard.

(2) The second source of difficulty level is calculated according to the item analysis.[5] The difficulty level of item analysis refers to the percentage of students who answered the question correctly in the testing environment. However, to remain consistent with the interpretation of other models, we use the percentage of students who answered the question incorrectly (Yeung, 2019).

(3) The third source of difficulty level is the question difficulty learned by PKT in modeling question solving. Considering the one-to-many and many-to-many relationships between questions and concepts, PKT generates different difficulty values for each question under different concepts, with difficulty values $\in [0, 1]$. For the purpose of comparison and analysis, we use a weighted average method to combine the difficulty values of the same question under different concepts as the final difficulty for the question. By utilizing a staged modeling approach to analyze the entire learning process and referencing the two-parameter logic model, PKT has learned more reliable question difficulty.

The Pearson correlation test results indicate that the difficulty levels learned from PKT are more strongly correlated with the other two sources (see Table 4).When compared with the first source of difficulty, which is publisher difficulty, the correlation coefficient between PKT's difficulty level and it is 0.24, while the correlation coefficient between the difficulty level learned by Deep-IRT and publisher difficulty is only 0.08. When compared with the second source of difficulty, Item Analysis Difficulty, the Pearson correlation coefficient between PKT's difficulty level and it is 0.78 (indicating a strong correlation), which is higher than the correlation coefficient of 0.56 (indicating moderate correlation) between the Deep-IRT model and the item analysis model. This strongly indicates that under the two standards of "Publisher Difficulty" and "Item Analysis Difficulty", the question difficulty learned by PKT is more reliable than that learned by the Deep-IRT model.[6]

While both PKT and Deep-IRT are inspired by IRT in generating question difficulty, PKT may have discovered more learning rules due to its modeling of the complete three-stage learning process. As a result, the difficulty level learned by PKT is more realistic and closely aligned with the actual difficulty of the questions.

We visualize the difficulty level obtained from the three sources (Publisher Difficulty, Item Analysis Difficulty, and PKT Difficulty) to explore their correlations. Fig. 5 displays box plots, histograms, scatter plots, and probability density function graphs describing the pairwise relationships among the three datasets.

Publisher difficulty only has three levels $\{1, 2, 3\}$, while the other two types of difficulty are distributed between 0 and 1, which may be why their correlation coefficients with publisher difficulty are lower. To better understand this issue, we create box plots of PKT difficulty and item analysis difficulty on the category axis of publisher difficulty levels. As subplots (4) and (7) in Fig. 5 show, both PKT difficulty and item analysis difficulty increase as publisher difficulty increases from left to right. This suggests that these two difficulties are indeed correlated with publisher difficulty, and it is the difference in data type that leads to the poor performance under the Pearson correlation coefficient test.

Subplot (6) in Fig. 5 shows the relationship between PKT difficulty and item analysis difficulty, with dots of three different colors representing the three levels of publisher difficulty. It is evident that (1) the dots exhibit a linear trend, distributed from the lower left to the upper right, indicating a correlation between the two types of difficulty; (2) the dots of the third color are mostly situated in the upper right and less in the lower left, suggesting that when item analysis difficulty and PKT difficulty are high, publisher difficulty is usually high as well. This finding indicates a certain degree of correlation and consistency between the difficulty of the questions derived from the three different sources.

In summary, the above tests and analyses indicate that the difficulty levels learned by PKT align with those of the two benchmarks and are more reliable than those learned by Deep-IRT.

(2) From a case analysis perspective: Examining the reliability of question difficulty

In this experiment, we use 948 questions from the Eedi platform released in the NeurIPS2020 Educational Data Mining Challenge for case analysis.

Fig. 6 shows example questions under two different concepts, "Multiplying and Dividing Negative Numbers" and "Multiples and Lowest Common Multiple". The last column shows the corresponding difficulty level learned by PKT for each question. For each concept, we list one difficult question, one moderate question, and one easy question. The difficulty value increases with question difficulty. Combining the information in each question, we can see that the learned difficulty levels closely match our understanding of question difficulty. Overall, the difficulty levels predicted by PKT closely correspond to the real difficulty levels of the questions.

These results demonstrate the potential of PKT to be applied in real-world educational environments, as the difficulty levels learned by PKT closely align with both publisher difficulty and the intuitive difficulty of questions in the real world.

The ability of PKT to predict question difficulty has many practical applications. (1) Applied to adaptive testing: dynamically adjusts question difficulty based on a student's performance, thereby better assessing their level; (2) Applied to recommending teaching resources: by tracing the knowledge concepts that students are capable of mastering, and predicting the question difficulty, recommends the appropriate teaching resources accordingly; (3)Applied to course design and teaching plan adjustments: helps course designers better design and adjust course content, improving teaching effectiveness by predicting and evaluating the difficulty levels of questions in the course.

5.4.2. *Visualizing difficulty distribution*

(1) Exploring difficulty distribution in datasets

PKT generates the difficulty value for each question, and the above tests and analysis demonstrate that these difficulty values are reliable. Furthermore, exploring whether the difficulty distribution of questions in the dataset is balanced is necessary to ensure that the questions cover a wide range of difficulty levels. If the questions are too easy or too difficult, the assessment may fail to differentiate between high

[5] https://www.washington.edu/assessment/scanning-scoring/scoring/reports/item-analysis/

[6] The correlation coefficient between Publisher Difficulty and Item Analysis Difficulty is 0.41, which is consistent with the 0.40 correlation degree between the two in the Deep-IRT paper.

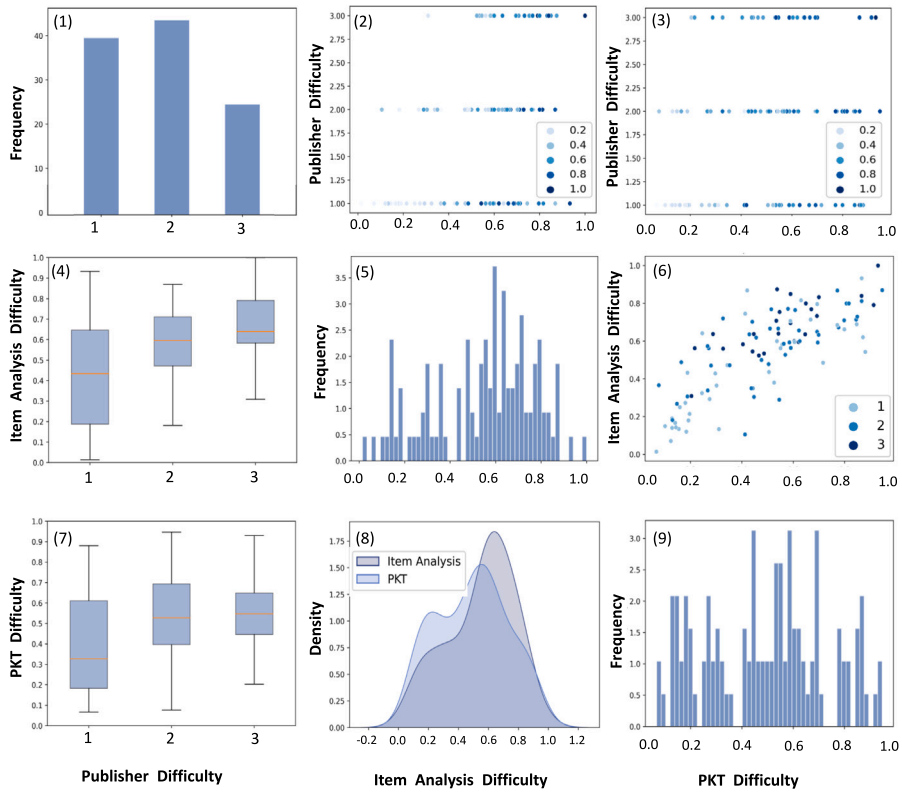

Fig. 5. Visual analysis of the relationship between three different types of difficulty obtained from various sources. Among them, publisher difficulty is a discrete value in $\{1, 2, 3\}$, item analysis difficulty and PKT difficulty are distributed between 0 and 1 more continuously.

concept	question	β
Multiplying and Dividing Negative Numbers	The product of two numbers is -30. What could the two numbers be? A. -15 and -2 B. -20 and -10 C. -6 and 5 D. -30 and 0	0.8838
	(-2) x (-8) + (-2) x 5 = (-2) x (□ + □) = □ What belongs in the boxes? A. -8 5 -6 B. -8 5 6 C. 8 5 -6 D. 8 -5 -6	0.3662
	21÷ (-3) = A.7 B.-7 C.24 D.-24	0.1482
Multiples and Lowest Common Multiple	In which region does 6 belong? Factors of 30 A B Multiples of 12 C D	0.8535
	Is the following statement always true, sometimes true, or never true? *One less than a multiple of 5 is a multiple of 4* A. Never true B. Sometimes true C. Always true D. Not enough information to decide	0.5305
	A common multiple of 5 and 6 is... A. 1 B. 11 C. 30 D. They do not have a common multiple	0.1358

Fig. 6. Question texts and the difficulty parameter β learned by PKT for selected questions on two concepts on the Eedi dataset. Learned difficulty levels match our intuitive understanding of the difficulty of these questions.

and low-performing students, leading to a reduction in the reliability and validity of the assessment.

To study the difficulty distribution in the six datasets used in our experiment, we use histograms to describe the distribution of question difficulty and fit a normal distribution curve. The statistical information of the six datasets can be found in Table 2, and the visualization of the distribution of question difficulty in the datasets is shown in Fig. 7.

Fig. 7 visualizes the distribution of question difficulty across six datasets. The exploration shows that the difficulty distribution of questions in the Assist09 and EdNet datasets used in our experiment is closer to a normal distribution, with a higher proportion of moderately difficult questions and fewer very easy or very difficult questions.

(2) Exploring difficulty distribution of multi-concept questions

Having explored the distribution of overall question difficulty in used datasets, we do a more refined exploration.

If a question corresponds to two or more concepts, we define it as a multi-concept question. One of our intuitions is that multi-concept questions are likely to be more difficult than single-concept questions, and if a question corresponds to more concepts, it may be more difficult. Therefore, we investigate the distribution of question difficulty under different numbers of concepts in Assist09 and Assist17 datasets to validate this hypothesis.

Table 5 presents information on the questions in the dataset that relate different numbers of concepts, including the mean and median

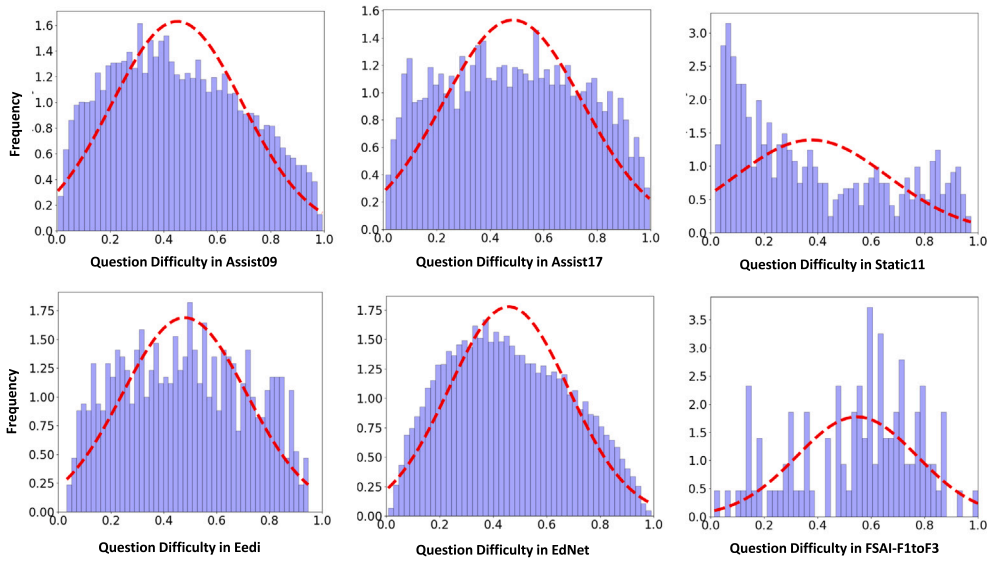

Fig. 7. Explore the distribution of question difficulty in the 6 datasets.

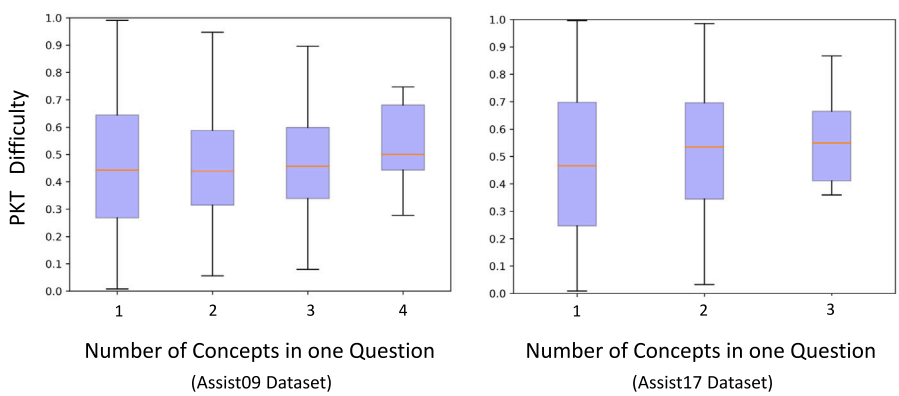

Fig. 8. On Assist09 and Assist17 dataset, as the number of concepts related in one question increases, the mean difficulty of the question increases and the box plot shifts upwards.

Table 5
Difficulty feature statistics of questions associating different numbers of concepts in Assist09 and Assist17 datasets.

Dataset	Number of concepts in one question	Average difficulty	Median difficulty	Number of questions
Assist09	1	0.45	0.44	10,041
	2	0.45	0.44	2,014
	3	0.47	0.46	277
	4	0.53	0.50	8
Assist17	1	0.47	0.46	2,467
	2	0.52	0.53	682
	3	0.55	0.55	16

difficulty values of the questions under the corresponding numbers of concepts, as well as the number of questions.

Upon observing the Assist09 and Assist17 dataset, we notice that the majority of questions are linked with a single concept, while the fewest questions are linked with four concepts. Furthermore, we discover that the average and median difficulty of a question increases as the number of associated concepts increases.

As shown in Fig. 8, from left to right, as the number of concepts increases, the box plot moves upward, indicating that the overall difficulty of the questions increases with the corresponding number of concepts. This confirms our intuition that the difficulty of a question is positively correlated with the number of concepts it is associated with.

6. Conclusion

We present Progressive Knowledge Tracing (PKT), an approach that thoughtfully simulates learning in stages: concept mastery, question-solving, and answering behavior. By incorporating elements from the two-parameter logistic model and educational theories, such as constructivism, PKT creates parameters to provide insights into student learning. Performance experiments across six datasets indicate that PKT performs favorably compared to baseline methods. It also shows promise in modeling question difficulty and discrimination with a higher degree of interpretability. For educators, PKT could be a helpful

tool in understanding students' concept mastery and pinpointing error sources, which may assist in making informed instructional adjustments. Intelligent tutoring systems might also find PKT useful for making more targeted resource recommendations.

As with many novel methods, PKT also has room for improvement. For instance, considering that students' actual answering behaviors may extend beyond mere guessing and slipping, exploring and modeling other behaviors, such as hesitation and repeated answer modifications, may offer new opportunities for model optimization. Additionally, studying students' question-solving strategies and providing a personalized assessment of question difficulty might aid in more accurately predicting students' responses. Moving forward, we are cautiously optimistic about integrating PKT into our Intelligent Tutoring System, aiming to enhance student outcomes. The interpretability of PKT and its capability to provide insights may facilitate better integration of teaching strategies with individual learning needs, potentially making a positive contribution to the field of knowledge tracing.

CRediT authorship contribution statement

Jianwen Sun: Supervision, Writing – original draft, Writing – review & editing. **Mengqi Wei:** Data curation, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing. **Jintian Feng:** Formal analysis, Investigation, Writing – review & editing. **Fenghua Yu:** Data curation, Investigation, Writing – review & editing. **Qing Li:** Conceptualization, Project administration, Writing – review & editing. **Rui Zou:** Conceptualization, Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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